Student Management System

➢ Introduction:

The program is a basic **Login System** implemented in C that supports two types of users: **students** and **teachers**. It requires users to enter their username and password to gain access. The credentials are checked a predefined list of users stored within the program. Upon successful login, the system identifies the user's role and provides a tailored to that role.

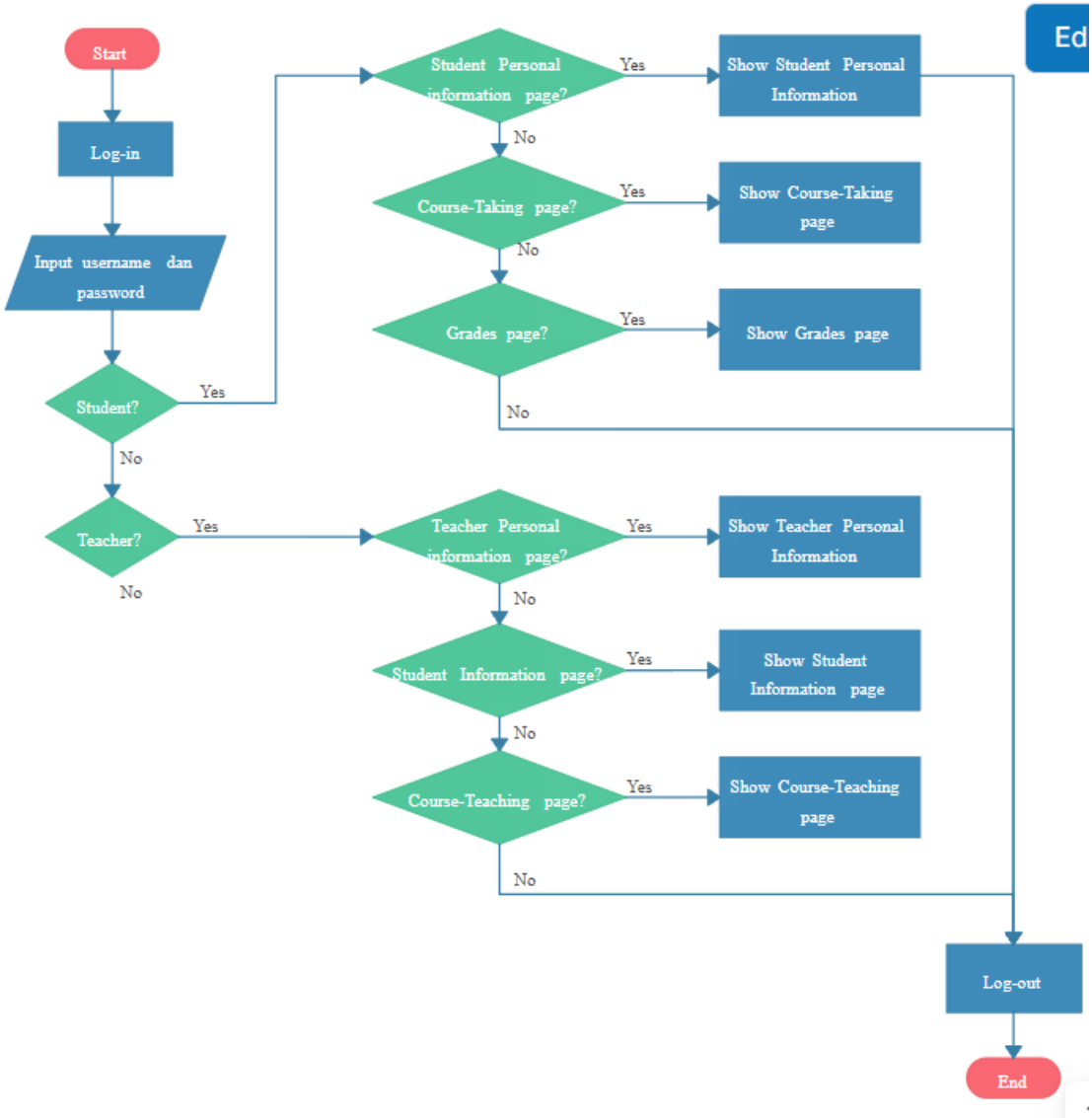
For students, the menu includes options to view their personal information,course-taking page, and grades page. Teachers have different menu with options to view their personal information, student information page, and course-teaching page. Users can interact with these menus until they decide to log out, which ends their session. This design demonstrates role-based access control and simple user interaction using console input and output.

Overall, this program illustrates important programming concepts such as user authentication, conditional branching, loops for menu-driven interaction, and modular code structure through separate functions for each role's menu. It serves as a useful foundation for understanding how to manage different user roles and build interactive console applications in C.

➢ Algorithm:

1) **Start the program.**
2) Login**:**
- **Prompt user to input username and password.**
3) **Check User Role:**
- **If user is Student, go to step 4.**
- **Else if user is Teacher, go to step 8.**
- **Else show invalid login message and go back to step 2 or end program.**
4) **Student Menu:**
- **Ask if user wants to see Student Personal Information page.**
    - **If Yes, show Student Personal Information page.**
- **Else ask if user wants to see Course-Taking page.**
    - **If Yes, show Course-Taking page.**
- **Else ask if user wants to see Grades page.**
    - **If Yes, show Grades page.**
- **After showing requested page, return to step 4 for another selection or log out.**
5) **Ask if user wants to Log out:**
- **If Yes, go to step 11.**
- **If No, return to step 4.**
6) **Teacher Menu:**
- **Ask if user wants to see Teacher Personal Information page.**
    - **If Yes, show Teacher Personal Information page.**
- **Else ask if user wants to see Student Information page.**
    - **If Yes, show Student Information page.**
- **Else ask if user wants to see Course-Teaching page.**
    - **If Yes, show Course-Teaching page.**
- **After showing requested page, return to step 6 for another selection or log out.**
7) **Ask if user wants to Log out:**
- **If Yes, go to step 11.**
- **If No, return to step 6.**
8) **Log out:**
- **Perform log out operation (clear session or confirm log out).**
9) **End the program.**

➢ FLOWCHART:


Start
Log-in
Input username dan password
Student?
Yes
No
Teacher?
Yes
No
Student Personal information page?
Yes
No
Show Student Personal Information
Course-Taking page?
Yes
No
Show Course-Taking page
Grades page?
Yes
No
Show Grades page
Teacher Personal information page?
Yes
No
Show Teacher Personal Information
Student Information page?
Yes
No
Show Student Information page
Course-Teaching page?
Yes
No
Show Course-Teaching page
Ed
Log-out
End

➢PROGRAM

```
#include <stdio.h>
#include <string.h>

// Simple hardcoded users for demo
struct User {
   char username[20];
   char password[20];
   char role[10];  // "student" or "teacher"
};

struct User users[] = {
   {"student1", "pass123", "student"},
   {"teacher1", "teach123", "teacher"}
};

int login(char *username, char *password, char *role) {
   for (int i = 0; i < 2; i++) {
      if (strcmp(username, users[i].username) == 0 && strcmp(password, users[i].password) == 0) {
         strcpy(role, users[i].role);
         return 1;
      }
   }
   return 0;
}

void studentMenu() {
   int choice;
   while (1) {
      printf("\n--- Student Menu ---\n");
      printf("1. Personal Information\n");
      printf("2. Course-Taking Page\n");
      printf("3. Grades Page\n");
      printf("4. Logout\n");
      printf("Enter choice: ");
      scanf("%d", &choice);

      switch(choice) {
```

```
        case 1:
            printf("Showing Student Personal Information...\n");
            break;
        case 2:
            printf("Showing Course-Taking Page...\n");
            break;
        case 3:
            printf("Showing Grades Page...\n");
            break;
        case 4:
            printf("Logging out...\n");
            return;
        default:
            printf("Invalid choice, try again.\n");
    }
  }
}

void teacherMenu() {
  int choice;
  while (1) {
    printf("\n--- Teacher Menu ---\n");
    printf("1. Personal Information\n");
    printf("2. Student Information Page\n");
    printf("3. Course-Teaching Page\n");
    printf("4. Logout\n");
    printf("Enter choice: ");
    scanf("%d", &choice);

    switch(choice) {
        case 1:
            printf("Showing Teacher Personal Information...\n");
            break;
        case 2:
            printf("Showing Student Information Page...\n");
            break;
        case 3:
            printf("Showing Course-Teaching Page...\n");
            break;
        case 4:
```

```
            printf("Logging out...\n");
            return;
        default:
            printf("Invalid choice, try again.\n");
    }
  }
}

int main() {
  char username[20], password[20], role[10];
  int loggedIn = 0;

  printf("=== Welcome to Login System ===\n");

  while (!loggedIn) {
    printf("Enter username: ");
    scanf("%s", username);
    printf("Enter password: ");
    scanf("%s", password);

    if (login(username, password, role)) {
      printf("Login successful! Role: %s\n", role);
      loggedIn = 1;
    } else {
      printf("Invalid username or password. Try again.\n");
    }
  }

  if (strcmp(role, "student") == 0) {
    studentMenu();
  } else if (strcmp(role, "teacher") == 0) {
    teacherMenu();
  }

  printf("=== End of Program ===\n");
  return 0;
}
```

➢OUTPUT

```
=== Welcome to Login System ===
Enter username: student1
Enter password: pass123
Login successful! Role: student

--- Student Menu ---
1. Personal Information
2. Course-Taking Page
3. Grades Page
4. Logout
Enter choice: 1
Showing Student Personal Information...

--- Student Menu ---
1. Personal Information
2. Course-Taking Page
3. Grades Page
4. Logout
Enter choice: 3
Showing Grades Page...

--- Student Menu ---
1. Personal Information
2. Course-Taking Page
3. Grades Page
4. Logout
Enter choice: 4
Logging out...
=== End of Program ===
```

THANK YOU